

Module 2

Linear Time-Invariant (LTI) Systems

Section A – Theory

Problem 1

Define a Linear Time-Invariant (LTI) system.

State the conditions required for a system to be:

- Linear
- Time-invariant

Problem 2

Determine whether the following systems are LTI or not.

(a) $y(t) = 2x(t)$

(b) $y(t) = x^2(t)$

(c) $y(t) = x(t) + t$

(d) $y(t) = x(t-3)$

Problem 3

State whether the following systems are:

- Stable
- Unstable

(a) $y(t) = 5x(t)$

(b) $y(t) = e^{x(t)}$

(c) $y(t) = \int_{-\infty}^t x(\tau) d\tau$

(d) $y[n] = 0.5x[n]$

Problem 4

Determine whether the following systems are invertible.

(a) $y(t) = 2x(t)$

(b) $y(t) = x^2(t)$

(c) $y(t) = x(t-1)$

(d) $y[n] = x[n] + 3$

Problem 5

Explain the significance of the impulse response of an LTI system.

Why is it sufficient to completely characterize an LTI system?

Section B – Numerical Problems

Problem 6

Find the convolution of

$$x[n] = \{1, 2, 1\}$$

$$h[n] = \{1, 1\}$$

Problem 7

Compute

$$y(t) = x(t) * h(t)$$

where $x(t) = u(t)$

and $h(t) = e^{-t}u(t)$

Problem 8

Find the convolution of $x[n] = \{2, 1\}$

and $h[n] = \{1, 2, 1\}$

Problem 9

Determine whether the following impulse responses correspond to stable systems.

(a) $h[n] = (0.5)^n u[n]$

(b) $h[n] = 2^n u[n]$

(c) $h[t] = e^{-2t} u[t]$

Problem 10

Verify the commutative property of convolution using

$x[n] = \{1, 2\}$ and $h[n] = \{3, 1\}$

Section C – Python Exercises

Problem 11

Generate and plot

- Unit impulse
- Unit step

Compute their discrete convolution.

Problem 12

Compute and plot the convolution of $x[n] = \{1, 2, 3\}$

and $h[n] = \{1, 1, 1\}$

Problem 13

Generate and plot

$h[n] = (0.5)^n u[n]$

Problem 14

Verify the commutative property using Python.

Compute $x * h$ and $h * x$

for $x[n] = \{1, 2, 3\}$

and $h[n] = \{2, 1\}$

Problem 15

Implement a moving-average filter.

Input signal:

$x[n] = \{2, 4, 6, 8, 10, 12\}$

Filter:

$h[n] = \frac{1}{3} \{1, 1, 1\}$

Plot both the input and filtered output.